

Feature Fusion Based Adversarial Example Detection Against Second-Round Adversarial Attacks

Abstract—Convolutional Neural Networks (CNNs) show strong performance across many fields especially in tasks involving pattern recognition and visual data. Still, their reliability can be challenged and threatened by adversarial examples. Adversarial examples are designed to mislead CNNs and make them to produce incorrect outputs. There are many methods that can be used to detect adversarial examples, however most detection based defense models are vulnerable to second-round adversarial attacks which are able to deceive the base model and the detector simultaneously. Handcrafted steganalysis features are used to counter second-round adversarial attacks, Spatial Rich Model (SRM) is the Handcrafted Steganalysis Feature used in this research. However, SRM works only in the spatial domain and achieves low accuracy when dealing with small perturbations. To overcome this, Wavelet features are added to capture perturbations in the frequency domain. This article proposes to combine SRM handcrafted features, wavelet features, and deep features of the CNN in a triple fusion approach to increase the detection accuracy and to counter second-round adversarial attacks.

1. Introduction

This work is based on the study proposed by Qin et al. [16].

Convolutional Neural Networks have become fundamental in many modern machine learning applications due to their strong performance in tasks such as image classification [1]-[3], object detection [4]-[6]. Despite having such strong sides they are vulnerable to adversarial examples which are specially designed to deceive the model and force wrong predictions [7] which are inputs that are intentionally modified to produce incorrect outputs. These modifications are usually too small to be noticed by the human eye.

Most of the adversarial attacks are based on minimizing the difference between the adversarial examples and the original images. To defend against these attacks, many methods have been proposed. While some approaches focus on making the model itself more robust some preprocess the input images before classification phase. These approaches are often effective to some extent but reduce accuracy.

Detection based defenses have become the preferred approach against adversarial attacks. These methods aim to identify if an input is adversarial before it reaches the classifier. However these detection based approaches are vulnerable to second-round-adversarial attacks, which are specially designed to deceive both the model and the detector at the same time.

To counter this issue, prior work utilized handcrafted steganalysis features such as the Spatial Rich Model (SRM) to detect such attacks. While being useful, this approach works only in the spatial domain and struggled with small modifications, achieving limited accuracy. To overcome this limitation, wavelet features are introduced to capture adversarial perturbations in the frequency domain, which the SRM alone fails to detect. This paper proposes to combine handcrafted steganalysis features (SRM), wavelet features, and deep features from CNNs in a triple fusion approach to improve detection accuracy and counter second-round adversarial attacks.

2. Related Works

A. Spatial Rich Model

SRM [8] is a handcrafted feature extraction method developed originally for steganalysis. SRM works by applying a large set of high-pass filters to an image to capture subtle noise changes that are invisible to the human eye. Since adversarial perturbations leave behind similar traces, Liu et al. [9] proposed using SRM features to detect adversarial examples. This method showed strong performance especially against second-round adversarial attacks. The non-differentiable operations in SRM makes it difficult for attackers. However, SRM struggles with large or sparse perturbations because of its internal quantization and truncation steps.

B. Second-Round Adversarial Attacks

Most detection-based defenses assume that the attacker is unaware of the detector. However, Carlini et al. [10] proposed second-round adversarial attacks, which target both the base model and the detector at the same time. The attacker treats the base model and the detector as a combined system and crafts examples that fool both simultaneously. This type of attack has been shown to be effective against many existing detectors.

C. Wavelet Transform

The Wavelet Transform [11] is a signal processing method that separates low-frequency structure from high-frequency details like edges and noise by decomposing the image into frequency sub-bands at multiple different scales. Since adversarial perturbations often appear as high frequency noise, Meenakshi and Maragatham [12] proposed using Discrete Wavelet Transform (DWT) features to detect adversarial examples by analyzing the high and mid frequency sub-bands where perturbations usually concentrate. This model captures small and subtle perturbations and its multi-scale nature complements spatial methods such as SRM.

3. Proposed Approach

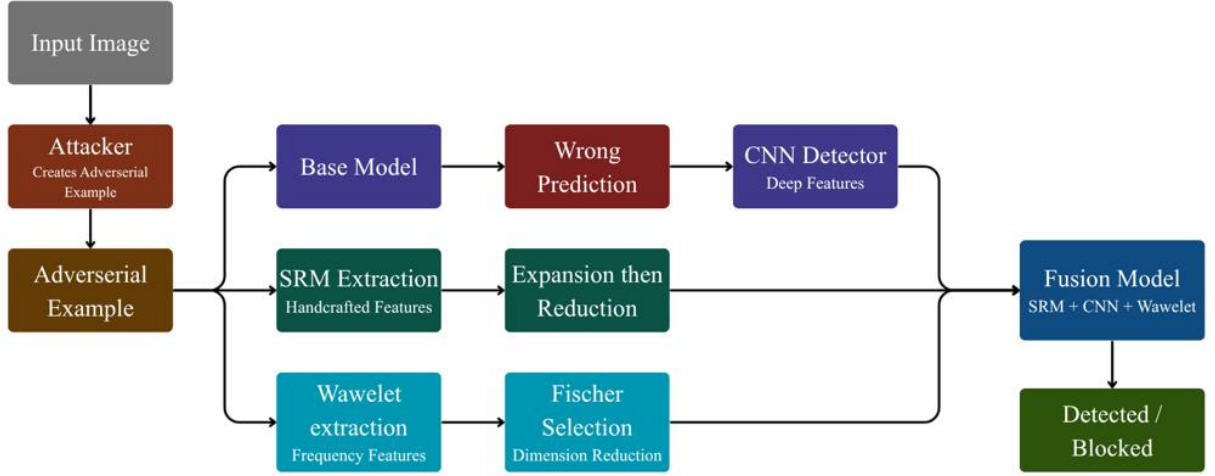


Fig. 1. Overview of the proposed fusion-based adversarial example detection framework.

The proposed approach builds on top of the already existing SRM + CNN fusion framework by adding a new extraction branch based on Discrete Wavelet Transform (DWT).

In this approach once an adversarial example is created it passes through 3 parallel pipelines at the same time.

The first pipeline feeds the image into the base model that produces a wrong prediction. Then the CNN detector extracts deep features from this process.

The second pipeline extracts SRM features by applying high-pass filters across each color channel of the image. Then Fisher score based selection is applied to keep only the most discriminative dimensions.

The third pipeline applies wavelet decomposition to the same image. Wavelet features are particularly sensitive to abrupt, localized changes in pixel intensity where SRM usually fails.

Finally all three feature sets are concatenated and passed into the fusion model. This fusion model makes the final decision.

4. Experimental Setup

1. Datasets and Base Models: Two widely studied datasets, CIFAR-10 [13] and Tiny-ImageNet[14], are utilized for performance evaluation in this article. ResNet18 [2] is adopted as the base model for both datasets, with the input stem adapted to the resolution of each dataset (32×32 for CIFAR-10 and 64×64 for Tiny-ImageNet). The base model is pretrained on the

training split of each dataset. For CIFAR-10, the test set contains 10,000 images, and for Tiny-ImageNet the validation set contains 10,000 images. These test images are used to construct the detector dataset, which will be used to train and evaluate the detector.

2. Adversarial Examples: The proposed detector aims to differentiate between natural images and adversarial examples, thus various adversarial attack methods are adopted to evaluate the detection ability of the proposed detector. For ℓ_∞ -based methods, we select PGD [15], which generates adversarial examples by iteratively applying small perturbations within a bounded range. PGD is applied with a perturbation budget of $\epsilon = 8/255$, a step size of $\alpha = 2/255$, and 20 iterations. To evaluate the robustness of the proposed model against second-round attacks, we adopt the method proposed by Carlini et al. [10]. This method generates adversarial examples that fool the base model and the detector at the same time by merging them as an N+1 classification model. The detector's adversarial score is converted into an additional logit and concatenated to the base model's output logits so that the generated example must force a wrong classification while avoiding being flagged as adversarial. This attack uses the same perturbation budget and step size as PGD but is run for 40 iterations this time. It has been proved to be effective against previous detectors, so we adopt it to evaluate the robustness of the proposed model against second-round attacks, which is also called adaptive attacks.

3. Training Process and Hyperparameters: The training process consists of two stages. In the first stage, the base model ResNet18 is trained using the Adam optimizer with a learning rate of $1e-3$ for 20 epochs, together with a cosine annealing learning rate schedule. ResNet18 is used for both CIFAR-10 and Tiny-ImageNet with input stem adapted to the resolution of each dataset. The batch size is set as 128 for CIFAR-10 and 64 for Tiny-ImageNet. In the second stage, the detector is trained using the Adam optimizer with a learning rate of $1e-3$ for 10 epochs. The weights of the base model are fixed in this stage. PGD is applied to the test split to generate adversarial examples and build the detector dataset. Natural images are labeled as 0 and adversarial examples are labeled as 1. The detector is then trained with natural image and corresponding adversarial example pairs because adversarial example detection is a binary classification task. SRM features are extracted from all images and Fisher score based feature selection is applied to reduce the dimensionality to 30. Discrete Wavelet Transform features are extracted from all images with using the Haar wavelet that captures frequency-level distortions across each color channel by computing the mean and standard deviation of each frequency sub-band, resulting in 24 dimensions. The deep features produced by the detector are used alongside the optimized SRM and wavelet features for the final triple fusion predictions. For comparison, four detector variants are trained under same settings: a standard CNN-only detector, a CNN+SRM fusion detector, a CNN + Wavelet fusion detector, and the proposed CNN + SRM + Wavelet triple fusion detector.

5. Performance Analysis

In this section we compare our proposed Triple Fusion model (CNN + SRM + Wavelet) with the SRM + CNN Fusion model alongside a plain CNN model and Wavelet + CNN Fusion model to evaluate the effect of adding wavelet features. Results are shown in the figures below:

A. CIFAR-10 Results

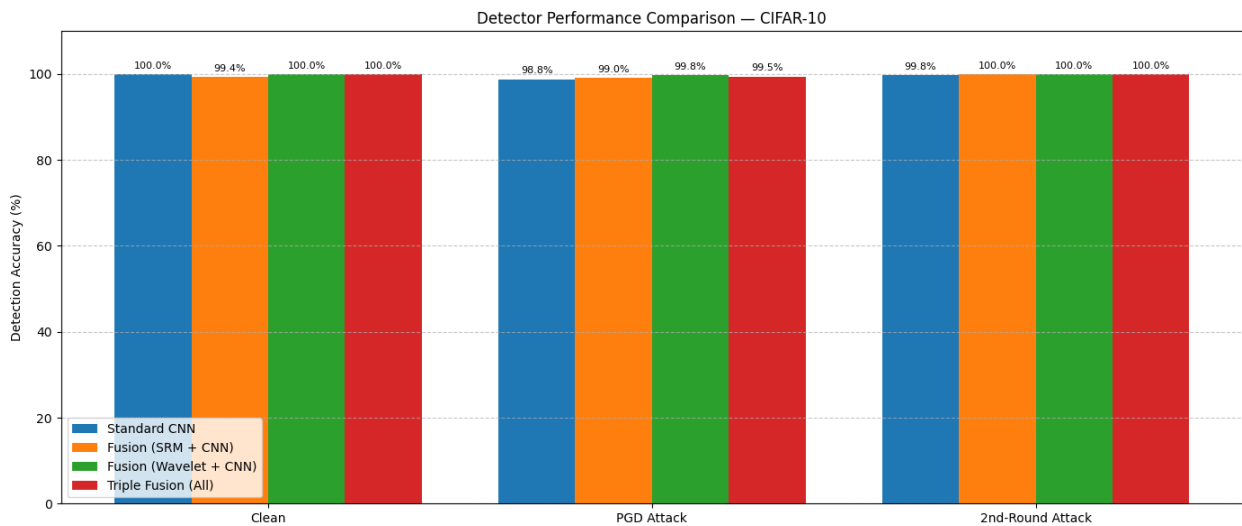


Fig. 2: First CIFAR10 Results.

First CIFAR-10 Results				
Metrics	Standard	SRM + CNN	Wavelet + CNN	Triple Fusion
Clean Accuracy	99.99%	99.41%	99.99%	100.00%
PGD Detection	98.82%	99.05%	99.77%	99.45%
2 nd -Round Detection	99.82%	99.95%	99.99%	99.95%

Table 1: First CIFAR-10 Results.

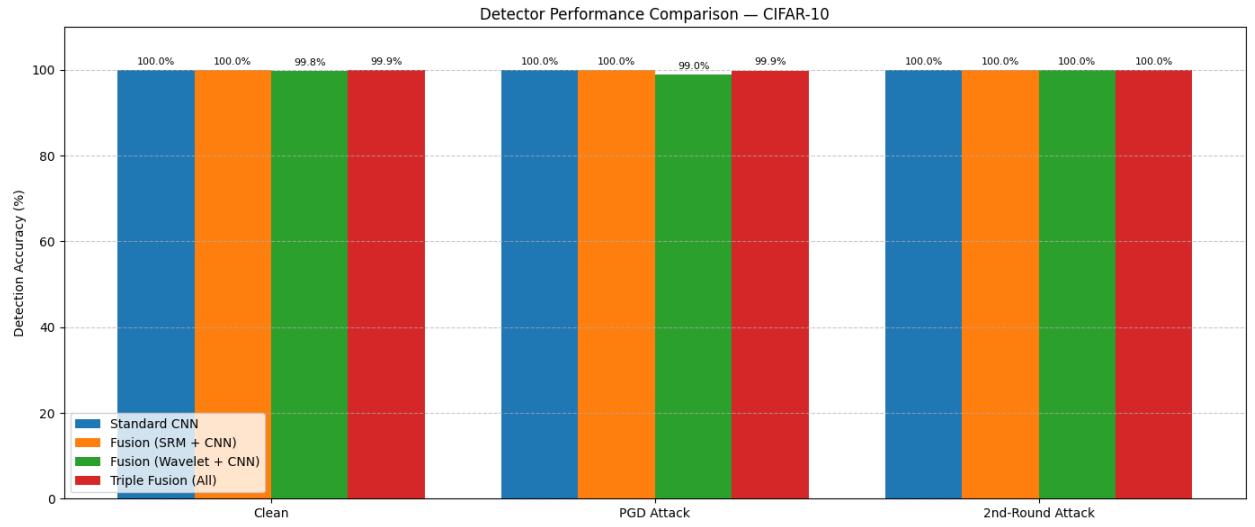


Fig. 3: Second CIFAR10 Results.

<i>Second CIFAR-10 Results</i>				
<i>Metrics</i>	<i>Standard</i>	<i>SRM + CNN</i>	<i>Wavelet + CNN</i>	<i>Triple Fusion</i>
<i>Clean Accuracy</i>	99.96%	100.00%	99.82%	99.90%
<i>PGD Detection</i>	99.99%	100.00%	99.00%	99.86%
<i>2nd-Round Detection</i>	100.00%	100.00%	100.00%	100.00%

Table 2: Second CIFAR-10 Results.

B. Tiny-ImageNet Results

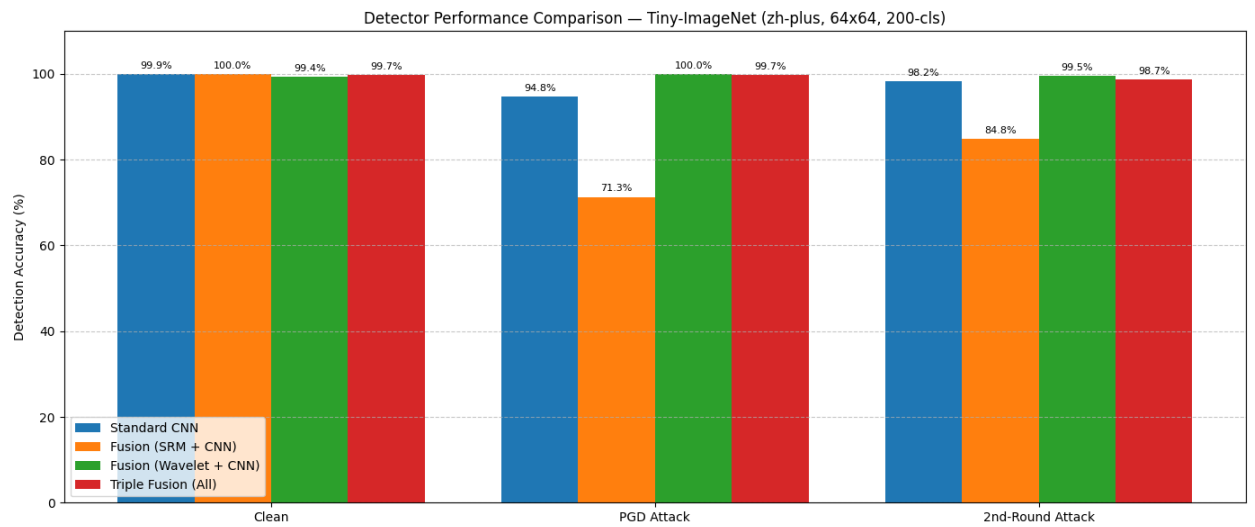


Fig. 4: First Tiny-ImageNet Results.

First Tiny-ImageNet Results

<i>Metrics</i>	<i>Standard</i>	<i>SRM + CNN</i>	<i>Wavelet + CNN</i>	<i>Triple Fusion</i>
<i>Clean Accuracy</i>	99.94%	99.97%	99.41%	99.68%
<i>PGD Detection</i>	94.80%	71.30%	99.99%	99.70%
<i>2nd-Round Detection</i>	98.21%	84.85%	99.48%	98.66%

Table 3: First Tiny-ImageNet Results.

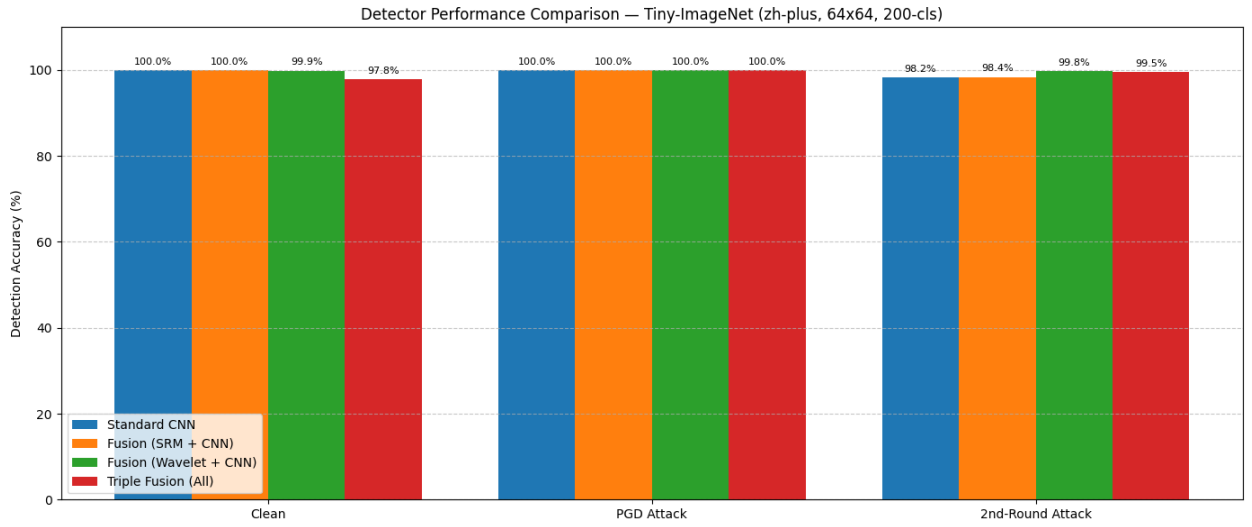


Fig. 5: Second Tiny-ImageNet Results.

Second Tiny-ImageNet Results

<i>Metrics</i>	<i>Standard</i>	<i>SRM + CNN</i>	<i>Wavelet + CNN</i>	<i>Triple Fusion</i>
<i>Clean Accuracy</i>	99.96%	99.97%	99.86%	97.85%
<i>PGD Detection</i>	99.97%	99.95%	99.99%	99.98%
<i>2nd-Round Detection</i>	98.22%	98.41%	99.79%	98.48%

Table 4: Second Tiny-ImageNet Results.

The four detectors are evaluated on CIFAR-10 and Tiny-ImageNet. Each experiment is repeated twice due to the randomness of the PGD attack initialization and to obtain more data for comparison. On CIFAR-10, all detectors perform similarly well with Clean Accuracy, PGD Detection and Second-Round Detection remaining close to 100% on both runs.

Tiny-ImageNet makes the differences appear more clearly. While SRM fusion detector (SRM+CNN) shows some instability, Wavelet Fusion (Wavelet + CNN) and Triple Fusion Detectors appear to be more stable.

6. Conclusion

This paper proposed a feature fusion based adversarial example detection framework that combined handcrafted SRM features, Wavelet based frequency features with deep features from a CNN base detector. Motivation behind this approach was that SRM, Wavelet and CNN detector complement each other. SRM captures spatial noise traces while wavelet features capture frequency level changes that SRM fails to detect. By combining all three, the proposed method addressed the limitations of each approach.

Experiments were conducted on two datasets using PGD as primary attack with a second-round attack proposed by Carlini et al. as the adaptive attack. Results show that proposed models achieved strong detection performance across both datasets.

Future works could explore applying this framework to more complex datasets and attack types.

7. References

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